

A FINAL REPORT
on
SYSTEMATIC EVALUATION OF CLUSTERING METHODS
FOR SCRNA-SEQ DATA

by

Shraddha Rajesh

2022A7PS0214U



BITS Pilani, Dubai Campus
Dubai International Academic City, Dubai
UAE

(Semester 1, 2025 – 2026)

ABSTRACT

Single-cell RNA sequencing (scRNA-seq) offers the capability of highly dimensional analysis of cellular heterogeneity but is of very high dimensionality, sparse, and highly technical noise. Dimensionality reduction is also, hence, a valuable quality clustering and biological interpretation requirement. It is a comparative analysis of four scRNA-seq clustering models (IGCLAPS, scCRT, scGCL and UIPBC); they are graph-based contrastive learning, transformer-based representation learning, and classical pipeline-based models. It is assessed by relying on publicly available scRNA-seq datasets on Gene Expression Omnibus (GEO) repository. The performance of clustering is tested by the Adjusted Rand Index (ARI) and the Normalized Mutual Information (NMI) and the performance of clustering is measured using Purity. The experimental results proved that learned representations, those that represent global dependencies, are more accurate, stable and robust in clustering compared to classical dimensionality reduction pipelines.

Key Words: scRNA-seq Data, Dimensionality Reduction, Transformer-based models, Graph-based Contrastive learning, Deep Learning, Clustering, Purity, Representation Learning.

ACKNOWLEDGEMENTS

It is true that without the help and advice of a few people, this study would not have been possible. I have been immensely pleased to have been given this chance to put my learning into practice by Dr. Souri Banerjee, Director of BITS Pilani, Dubai Campus through this meaningful project. I also wish to express my gratitude to Dr. Arumugam Somasundaram who was the Dean of AUGSD when I was registering the course. I would also like to profoundly express gratitude to my supervisor and mentor Dr. Hussain Chowdhury who is constantly supportive, wise and gives constructive criticism. The mentorship was important in making the plans in this project and helping me sail through all the phases with a lot of confidence and clarity.

TABLE OF CONTENTS

<i>Abstract</i>	2
<i>Acknowledgements</i>	3
1. Introduction	5
2. Aim & Objectives	6
2.1 Aim of the Work	6
2.2 Objectives	6
3. Background Study	7
3.1 Classical Pipeline-Based Approaches	7
3.2 Graph-Based Contrastive Learning	7
3.3 Transformer-Based Representation Learning	7
3.4 Methods	7
3.5 Datasets	8
4. Methodology	9
4.1 Data Preprocessing	9
4.2 Clustering	9
4.3 Evaluation Metrics	9
5. Experimental Results	10
5.1 Results	10
5.2 Average Performance Analysis	11
6. Key Insights	12
7. Conclusion & Future Scope	13
<i>References</i>	14

CHAPTER 1 – INTRODUCTION

Single-cell RNA sequencing has come to be a useful method of studying the cellular diversity in complicated biological systems. In comparison to bulk RNA sequencing, scRNA-seq has the ability to scale the amount of genes expressed by each cell, and in most applications data sets of tens of thousands of genes per cell are obtained. In spite of the richness of this, biology applications have been made fine-grained, but several challenges are posed.

First of all, scRNA-seq data are high-dimensional and distance-based clustering cannot be relied upon due to the curse of dimensionality. Second, the data is sparse, as the majority of the genes have zero or close-zero levels of the expression due to technical dropout. Then, the noise and batch effects make the process of retrieving meaningful patterns further complicated. The issues demand effective dimensionality reduction techniques that will be capable of preserving biological meaningful structure, albeit with the noise minimized.

Principal Component Analysis (PCA) is the most widespread linear dimensionality reduction methods used in the classical combinations of scRNA-seq clustering pipelines. However, the majority of the recent developments in the field of deep learning have provided representation learning algorithms that aim to learn more expressive embeddings. To be familiar with the impacts of these approaches on the performance of clustering, this paper compares and contrasts some of them.

In the unsupervised learning activity, dimensionality reduction is one of the main roles in clustering. The object of scRNA-seq analysis is to project the initial high-dimensional gene expression into a lower-dimensional form which captures significant biological variation.

CHAPTER 2 – AIM & OBJECTIVES

2.1 Aim of the Work:

- The objective of the present project is to examine the effect that different dimensionality reduction and deep learning methods have on the clustering performance of scRNA-seq data.

2.2 Objectives:

- To test and implement different scRNA-seq clustering models on different learning models.
- To evaluate the clustering performance of traditional biological measures of clustering.
- To evaluate the strength, steadiness and sound control features between datasets.
- To identify valuable information to select dimensionality reduction methods in scRNA-seq analysis.

CHAPTER 3 – BACKGROUND STUDY

Dimensionality reduction plays a central role in unsupervised learning tasks such as clustering. In scRNA-seq analysis, the goal is to reduce the original high-dimensional gene expression matrix into a lower-dimensional representation that preserves meaningful biological variation.

3.1 Classical Pipeline-Based Approaches:

- Pipeline-based algorithms PCA is usually used to reduce the dimensions, after which the clustering is done with an algorithm, like k-means or hierarchical clustering.
- These methods are efficient in terms of computation and simple to understand as well but may be incapable of finding complex non-linear relationships between genes.

3.2 Graph-Based Contrastive Learning:

- Unlike contrastive learning that generates a fixed number of samples, graph-based contrastive learning is unsupervised.
- Graph-based models form a graph in which the nodes represent cells and the edges the relationships of similarity.
- A contrastive learning method is then applied to attract similar cells and to repel dissimilar ones. Such techniques find a special purpose in dealing with sparsity and local structure.

3.3 Transformer-Based Representation Learning:

- Transformer models use a mechanism called self-attention to identify an interaction between genes in a global way.
- Unlike graph-based models, transformers do not deal with local neighbourhoods, and capture global dependencies more efficiently, which in turn produces more robust representations.

3.4 Methods:

- IGCLAPS:
 - IGCLAPS is a graph-based contrastive learning framework that constructs a cell–cell graph and learns representations by contrasting positive and negative graph views.
 - The method emphasizes local neighborhood consistency and aims to improve robustness against noise by enforcing representation similarity among biologically similar cells.

- **scCRT:**
 - scCRT is a transformer based model that uses self attention to scRNA-seq data to learn global gene gene and gene cell dependencies.
 - ScCRT unlike graph-based approaches does not only use local neighborhoods but instead can capture long-range interactions resulting in more expressive and stable representations.
- **scGCL:**
 - scGCL is a graph contrastive learning method which is aimed at learning strong cell-cell embeddings.
 - The scGCL can be sensitive to noise by using contrastive objectives, but its performance depends on the way the graph is built, as well as the sparsity of the underlying dataset.
- **UIPBC:**
 - UIPBC is a traditional pipeline based clustering methodology that involves dimensionality reduction using PCA and subsequent clustering and assessment.
 - It is a robust starting point and displays popular methods of scRNA-seq analysis.

3.5 Datasets:

- The datasets used in this project were obtained from the Gene Expression Omnibus (GEO) repository.
- A total of nine publicly available scRNA-seq datasets were selected, varying in number of cells, genes and cluster complexity.
- This diversity ensures that comparisons in performance reflect general trends rather than dataset-specific behavior.

S.No	Name	Cells	Genes	Clusters
1	GSE45719	291	45686	14
2	GSE48968	1378	45686	5
3	GSE52529	288	65218	3
4	GSE63818	328	65218	5
5	GSE64016	460	65218	4
6	GSE71585	749	45686	18
7	GSE75140	729	35238	13
8	GSE81076	1152	65218	12
9	GSE57872	864	65218	9

CHAPTER 4 – METHODOLOGY

4.1 Data Preprocessing:

- Standard preprocessing was performed on each dataset, including normalizing the data, filtering out the low-quality genes and cells, as well as feature selection wherever required.
- Pre-processing was done to meet the input needs of both approaches without making any unfair comparisons.

4.2 Clustering:

- Each method created low-dimensional embeddings based on the dimensionality reduction methods used.
- Clustering was then performed on the embeddings using either the default or recommended strategies for clustering, that is associated with each framework.

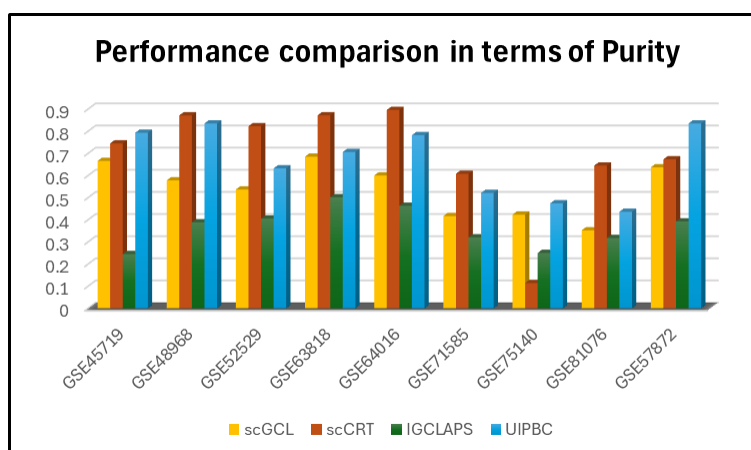
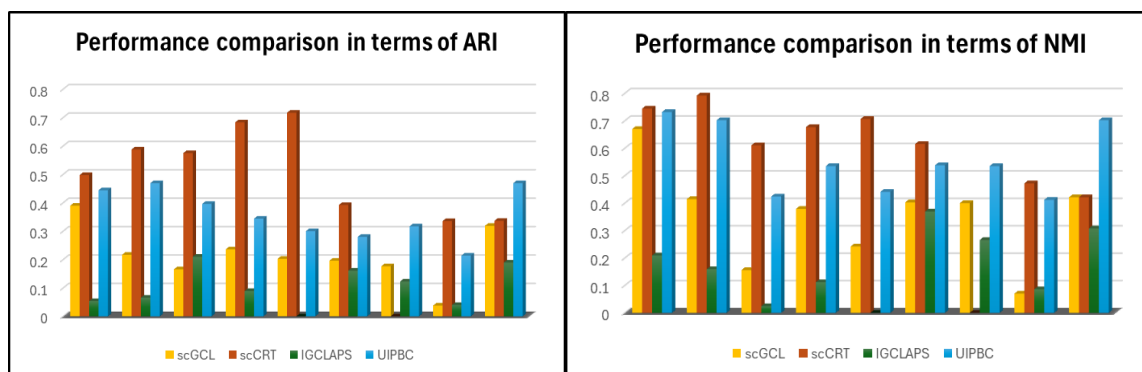
4.3 Evaluation Metrics:

- Comparison of clustering outcomes was done using three common measures:
 - *Adjusted Rand Index (ARI)*:
 - It measures the similarity between the predicted labels as well as true labels, while correcting for chance.
 - *Normalized Mutual Information (NMI)*:
 - It measures overlapping information of the predicted and the true cluster assignments.
 - *Purity*:
 - It measures the cluster homogeneity by assigning each cluster to its dominant true label.
- All these measures are used to evaluate accuracy, retention of information and the size of the clusters.

CHAPTER 5 – EXPERIMENTAL RESULTS

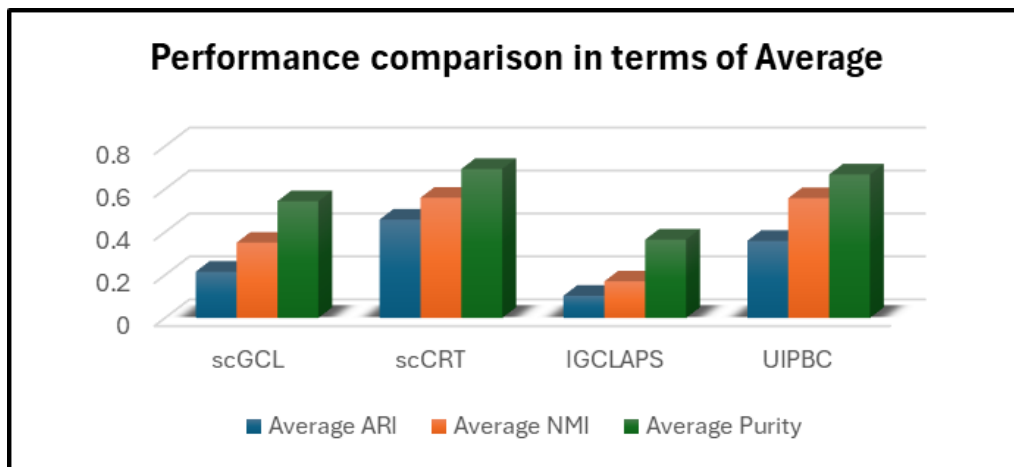
5.1 Results:

- Across the nine GEO datasets, transformer-based as well as graph-based methods consistently outperformed the classical pipeline.
- The highest performing model, i.e., scCRT was able to perform best on almost 7 out of 9 datasets and scored:
 - ARI ~ 0.73
 - NMI ~ 0.80
 - Purity ~ 0.90
- The graph based algorithms such as IGCLAPS and scGCL performed moderately, with ARI typically between 0.15-0.35, whereas UIPBC recorded constant but lower ARI values amongst the datasets.



5.2 Average Performance Analysis:

- In order to compare the general strength, the mean measures have been calculated on all datasets on each approach:
 - Average ARI:
 - Highest for the scCRT method.
 - Average NMI:
 - Always greater with learned representations compared to PCA.
 - *Average Purity*:
 - Transformer-based approach achieved the highest homogeneity in clustering.



CHAPTER 6 – KEY INSIGHTS

The experimental results clearly show that representation learning plays a more vital role than the clustering algorithm itself. Among all methods, the transformer-based approach consistently showcased superior performance, both with respect to peak metrics and stability across all the datasets.

The graph-based contrastive methods improved robustness as compared to classical pipelines but still exhibited higher sensitivity when it came to sparsity or the graph construction quality. UIPBC, while stable, lacked the expressive capacity needed to capture the complex nonlinear structure of data.

The transformer-based model, however, performed the best as it captures global dependencies, while reducing the reliance on local neighborhood relationships, and produces stable embeddings across heterogenous datasets. Stability across datasets emerged as a more reliable indicator of real-world usability than isolated peak performance.

CHAPTER 7 – CONCLUSION & FUTURE SCOPE

This work presents a comprehensive comparative evaluation of dimensionality reduction techniques for scRNA-seq clustering. The results demonstrate that learned representations significantly enhance clustering accuracy, robustness as well as consistency.

While classical pipelines remain useful as baselines, advanced representation learning methods, especially those capturing global gene patterns, offer clear advantages in noise handling and sparsity. The project includes useful recommendations that can be used to choose dimensionality reduction approaches in scRNA-seq analysis.

The future work will be centered on incorporating the most successful strategy i.e. scCRT into the UIPBC pipeline, substituting the classical PCA-based dimensionality reduction with the current downstream clustering and evaluation system. The idea of this integration is to enhance clustering strength without distorting the analysis processes that are in place. The subsequent extensions that will be done are the use of biological annotations, pathway analysis and downstream functional interpretation.

REFERENCES

- [1] Chowdhury, H. A. et al. (2022). UIPBC: an effective clustering for scRNA-seq data analysis without user input. *Knowledge-Based Systems*, 248.
- [2] Xiong, Z. et al. (2023). scGCL: an imputation method for scRNA-seq data based on graph contrastive learning. *Bioinformatics*, 39(3).
- [3] Zheng, W. et al. (2025). IGCLAPS: an interpretable graph contrastive learning method with adaptive positive sampling for scRNA-seq data analysis. *Bioinformatics*, 41(8).
- [4] Shi, Y. et al. (2024). scCRT: a contrastive-based dimensionality reduction model for scRNA-seq trajectory inference. *Briefings in Bioinformatics*, 25(3).
- [5] Q. Deng, D. Ramsköld, B. Reinius, and R. Sandberg, “Single-cell RNA-seq reveals dynamic, random monoallelic gene expression in mammalian cells,” *Science*, vol. 343, no. 6167, pp. 193–196, Jan. 2014.
- [6] A. K. Shalek, R. Satija, J. Shuga, J. J. Trombetta, *et al.*, “Single-cell RNA-seq reveals dynamic paracrine control of cellular variation,” *Nature*, vol. 510, no. 7505, pp. 363–369, Jun. 2014.
- [7] C. Trapnell, D. Cacchiarelli, J. Grimsby, P. Pokharel, *et al.*, “The dynamics and regulators of cell fate decisions are revealed by pseudotemporal ordering of single cells,” *Nature Biotechnology*, vol. 32, no. 4, pp. 381–386, Apr. 2014.
- [8] F. Guo, L. Yan, H. Guo, L. Li, *et al.*, “The transcriptome and DNA methylome landscapes of human primordial germ cells,” *Cell*, vol. 161, no. 6, pp. 1437–1452, Jun. 2015.
- [9] N. Leng, L. F. Chu, C. Barry, Y. Li, *et al.*, “Oscope identifies oscillatory genes in unsynchronized single-cell RNA-seq experiments,” *Nature Methods*, vol. 12, no. 10, pp. 947–950, Oct. 2015.
- [10] B. Tasic, V. Menon, T. N. Nguyen, T. K. Kim, *et al.*, “Adult mouse cortical cell taxonomy revealed by single-cell transcriptomics,” *Nature Neuroscience*, vol. 19, no. 2, pp. 335–346, Feb. 2016.
- [11] J. G. Camp, F. Badsha, M. Florio, S. Kanton, *et al.*, “Human cerebral organoids recapitulate gene expression programs of fetal neocortex development,” *Proceedings of the National Academy of Sciences*, vol. 112, no. 51, pp. 15672–15677, Dec. 2015.
- [12] D. Grün, M. J. Muraro, J. C. Boisset, K. Wiebrands, *et al.*, “De novo prediction of stem cell identity using single-cell transcriptome data,” *Cell Stem Cell*, vol. 19, no. 2, pp. 266–277, Aug. 2016.
- [13] A. P. Patel, I. Tirosh, J. J. Trombetta, A. K. Shalek, *et al.*, “Single-cell RNA-seq highlights intratumoral heterogeneity in primary glioblastoma,” *Science*, vol. 344, no. 6190, pp. 1396–1401, Jun. 2014.